

Intraquery Parallelism

1 Overview

Intraquery parallelism refers to the execution of a single query in parallel on multiple processors and disks. While interquery parallelism focuses on throughput (handling many small queries), intraquery parallelism is essential for **speeding up long-running, complex queries**.

Parallel Evaluation: Sorting

Consider a relation partitioned by range across multiple disks.

- **Process:** Each partition is sorted in parallel on its local processor.
- **Consolidation:** The sorted partitions are simply concatenated to produce the final result.

2 Two Forms of Intraquery Parallelism

Execution of a single query can be parallelized in two distinct, yet complementary ways:

2.1 Intraoperation Parallelism

Focuses on speeding up **individual operations** within a query (e.g., sort, select, project, join).

- **Scalability:** Generally scales better as it handles large numbers of tuples.
- **Focus:** Executing a single operator (like a Join) across all available processors.

2.2 Interoperation Parallelism

Focuses on executing **different operations** in a query expression tree in parallel.

- **Independent Parallelism:** Evaluating operations that do not depend on one another simultaneously.
- **Pipelined Parallelism:** The output of one operation is streamed as input to another. Both execute in parallel on separate processors (Producer/Consumer model).

3 Execution and Architecture Models

The choice of algorithms for parallelizing query evaluation depends largely on the underlying machine architecture.

Shared-Nothing as the Base Model

In technical discussions, a **shared-nothing architecture** is often used as the base model because:

- It explicitly describes data transfers between processors.
- Algorithms designed for shared-nothing can be easily simulated or optimized for **shared-memory** (via pointers/shared buffers) or **shared-disk** systems.

3.1 Model Assumptions

To simplify algorithmic presentation, we typically assume:

- n processors $(P_0, \dots, P_{n-1})$ and n disks $(D_0, \dots, D_{n-1})$.
- Each processor P_i is directly associated with disk D_i .
- In real systems, a processor might control a set of disks, but the logical principles remain the same.

Intraoperation and interoperation parallelism are often used simultaneously to achieve maximum performance gain for analytical queries.